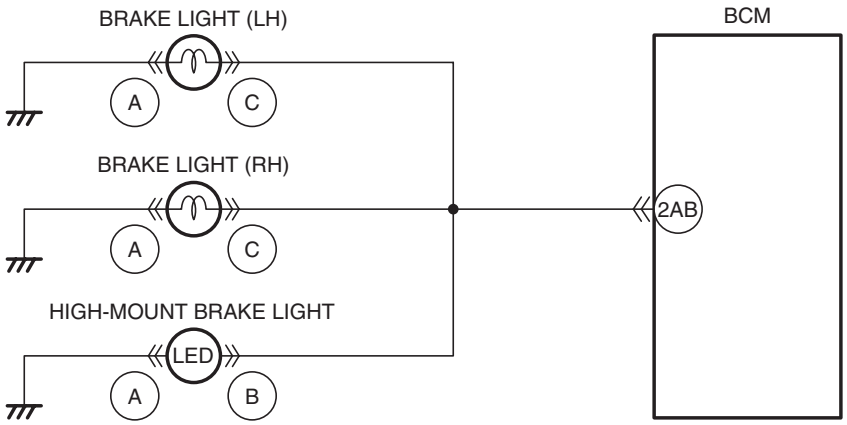
DTC C0023:14 [BCM]

id0902f5070600

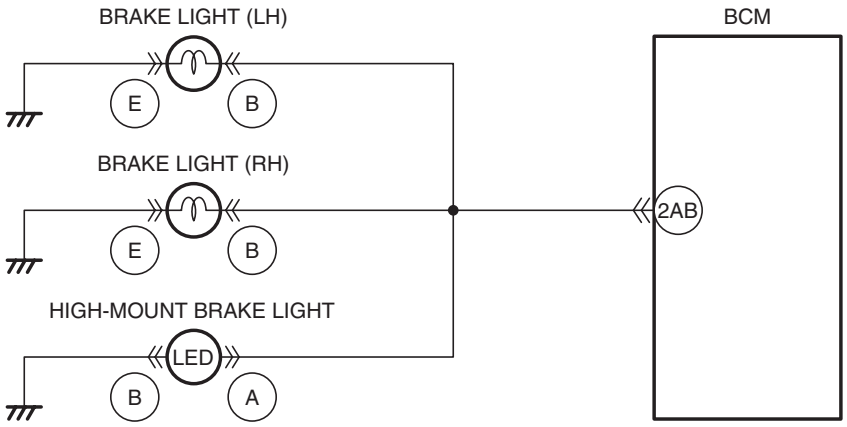
Description	Brake light circuit malfunction
Detection condition	• BCM detects any of the following conditions: — Brake lights/high-mount brake light illuminated even though driver or system does not illuminate brake lights/high-mount brake lights. — Brake lights/high-mount brake light not illuminated even though driver depresses brake pedal. — Brake lights/high-mount brake light not illuminated even though system illuminates brake lights/high-mount brake light.
Fail-safe	• Turn the brake light off.
Possible cause	• Brake light bulb (LH) malfunction • Brake light bulb (RH) malfunction • High-mount brake light connector or terminal malfunction • Brake light (LH) connector or terminal malfunction • Brake light (RH) connector or terminal malfunction • BCM connector or terminal malfunction • Short to ground in wiring harness between the following terminals: — 4SD: • Brake light (LH) terminal C—BCM terminal 2AB • Brake light (RH) terminal C—BCM terminal 2AB • High-mount brake light terminal B—BCM terminal 2AB — 5HB: • Brake light (LH) terminal B—BCM terminal 2AB • Brake light (RH) terminal B—BCM terminal 2AB • High-mount brake light terminal A—BCM terminal 2AB • Short to power supply in wiring harness between the following terminals: — 4SD: • Brake light (LH) terminal C—BCM terminal 2AB • Brake light (RH) terminal C—BCM terminal 2AB • High-mount brake light terminal B—BCM terminal 2AB — 5HB: • Brake light (LH) terminal B—BCM terminal 2AB • Brake light (RH) terminal B—BCM terminal 2AB • High-mount brake light terminal A—BCM terminal 2AB • Open circuit in wiring harness between the following terminals: — 4SD: • Brake light (LH) terminal C—BCM terminal 2AB • Brake light (RH) terminal C—BCM terminal 2AB • High-mount brake light terminal B—BCM terminal 2AB — 5HB: • Brake light (LH) terminal B—BCM terminal 2AB • Brake light (RH) terminal B—BCM terminal 2AB • High-mount brake light terminal A—BCM terminal 2AB • High-mount brake light malfunction • Brake light (LH) malfunction • Brake light (RH) malfunction • BCM malfunction

Description	Brake light circuit malfunction

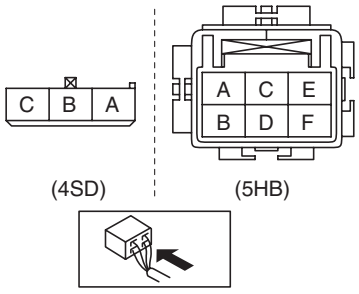
4SD



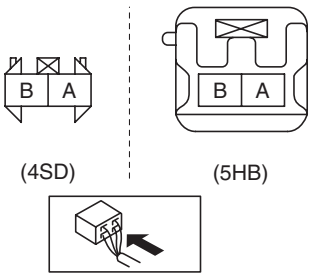
5HB



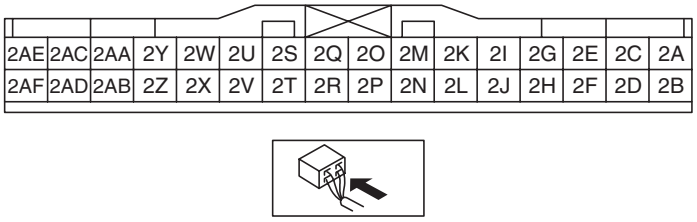
BRAKE LIGHT (LH)/(RH)
WIRING HARNESS-SIDE CONNECTOR



HIGH-MOUNT BRAKE LIGHT
WIRING HARNESS-SIDE CONNECTOR



BCM
WIRING HARNESS-SIDE CONNECTOR



Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY REAR BODY CONTROL MODULE (FBCM) DTCs AGAIN** • Clear the DTC for the BCM using the M-MDS. (See CLEARING DTC [BCM].) • Depress the brake pedal. • Retrieve the BCM DTCs using the M-MDS. (See DTC INSPECTION [BCM].) • Is the same DTC displayed?	Yes	**With rear combination light (bulb type):** • Go to the next step. **With rear combination light (LED type):** • Go to Step 4.
		No	Go to Step 18.
2	**VISUALLY INSPECT BRAKE LIGHT BULB (LH)** • Visually inspect the brake light bulb (LH) • Is the brake light bulb (LH) normal?	Yes	Go to the next step.
		No	Replace the brake light bulb (LH), then go to Step 17. (See BRAKE LIGHT/TAILLIGHT BULB REMOVAL/ INSTALLATION.)
3	**VISUALLY INSPECT BRAKE LIGHT BULB (RH)** • Visually inspect the brake light bulb (RH) • Is the brake light bulb (RH) normal?	Yes	Go to the next step.
		No	Replace the brake light bulb (RH), then go to Step 17. (See BRAKE LIGHT/TAILLIGHT BULB REMOVAL/ INSTALLATION.)
4	**INSPECT HIGH-MOUNT BRAKE LIGHT CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the high-mount brake light connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 17.
5	**INSPECT BRAKE LIGHT (LH) CONNECTOR CONDITION** • Disconnect the brake light (LH) connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 17.
6	**INSPECT BRAKE LIGHT (RH) CONNECTOR CONDITION** • Disconnect the brake light (RH) connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 17.
7	**INSPECT BCM CONNECTOR CONDITION** • Disconnect the BCM connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 17.

Step	Inspection		Action
8	**INSPECT BRAKE LIGHT CIRCUIT FOR SHORT TO GROUND** • Verify that the high-mount brake light, brake light (LH), brake light (RH) and BCM connectors are disconnected. • Inspect for continuity between BCM terminal 2AB (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • 4SD: — Brake light (LH) terminal C—BCM terminal 2AB — Brake light (RH) terminal C—BCM terminal 2AB — High-mount brake light terminal B—BCM terminal 2AB • 5HB: — Brake light (LH) terminal B—BCM terminal 2AB — Brake light (RH) terminal B—BCM terminal 2AB — High-mount brake light terminal A—BCM terminal 2AB **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 17.
		No	Go to the next step.
9	**INSPECT BRAKE LIGHT CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the high-mount brake light, brake light (LH), brake light (RH) and BCM connectors are disconnected. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off or on). • Measure the voltage at the BCM terminal 2AB (wiring harness-side). • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • 4SD: — Brake light (LH) terminal C—BCM terminal 2AB — Brake light (RH) terminal C—BCM terminal 2AB — High-mount brake light terminal B—BCM terminal 2AB • 5HB: — Brake light (LH) terminal B—BCM terminal 2AB — Brake light (RH) terminal B—BCM terminal 2AB — High-mount brake light terminal A—BCM terminal 2AB **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 17.

Step	Inspection		Action
10	**INSPECT BRAKE LIGHT CIRCUIT FOR OPEN CIRCUIT** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Verify that the high-mount brake light, brake light (LH), brake light (RH) and BCM connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — 4SD: • Brake light (LH) terminal C—BCM terminal 2AB • Brake light (RH) terminal C—BCM terminal 2AB • High-mount brake light terminal B—BCM terminal 2AB — 5HB: • Brake light (LH) terminal B—BCM terminal 2AB • Brake light (RH) terminal B—BCM terminal 2AB • High-mount brake light terminal A—BCM terminal 2AB • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • 4SD: — Brake light (LH) terminal C—BCM terminal 2AB — Brake light (RH) terminal C—BCM terminal 2AB — High-mount brake light terminal B—BCM terminal 2AB • 5HB: — Brake light (LH) terminal B—BCM terminal 2AB — Brake light (RH) terminal B—BCM terminal 2AB — High-mount brake light terminal A—BCM terminal 2AB **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 17.
11	**INSPECT HIGH-MOUNT BRAKE LIGHT FOR SHORT TO GROUND** • Verify that the high-mount brake light, brake light (LH), brake light (RH) and BCM connectors are disconnected. • Connect the high-mount brake light connector. • Inspect for continuity between BCM terminal 2AB (wiring harness-side) and body ground. • Is there continuity?	Yes	Replace the high-mount brake light, then go to Step 17. (See HIGH-MOUNT BRAKE LIGHT REMOVAL/ INSTALLATION.)
		No	Go to the next step.
12	**INSPECT HIGH-MOUNT BRAKE LIGHT FOR SHORT TO POWER SUPPLY** • Verify that the brake light (LH), brake light (RH) and BCM connectors are disconnected. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off or on). • Measure the voltage at the BCM terminal 2AB (wiring harness-side). • Is the voltage **0 V**?	Yes	**With rear combination light (bulb type):** • Go to Step 17. **With rear combination light (LED type):** • Go to the next step.
		No	Replace the high-mount brake light, then go to the next step. (See HIGH-MOUNT BRAKE LIGHT REMOVAL/ INSTALLATION.)
13	**INSPECT BRAKE LIGHT (LH) FOR SHORT TO GROUND** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Verify that the brake light (LH), brake light (RH) and BCM connectors are disconnected. • Connect the brake light (LH) connector. • Inspect for continuity between BCM terminal 2AB (wiring harness-side) and body ground. • Is there continuity?	Yes	Replace the rear combination light (LH), then go to Step 17. (See REAR COMBINATION LIGHT REMOVAL/ INSTALLATION.)
		No	Go to the next step.

Step	Inspection		Action
14	**INSPECT BRAKE LIGHT (LH) FOR SHORT TO POWER SUPPLY** • Verify that the brake light (RH) and BCM connectors are disconnected. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off or on). • Measure the voltage at the BCM terminal 2AB (wiring harness-side). • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Replace the rear combination light (LH), then go to Step 17. (See REAR COMBINATION LIGHT REMOVAL/ INSTALLATION.)
15	**INSPECT BRAKE LIGHT (RH) FOR SHORT TO GROUND** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Verify that the brake light (RH) and BCM connectors are disconnected. • Connect the brake light (RH) connector. • Inspect for continuity between BCM terminal 2AB (wiring harness-side) and body ground. • Is there continuity?	Yes	Replace the rear combination light (RH), then go to Step 17. (See REAR COMBINATION LIGHT REMOVAL/ INSTALLATION.)
		No	Go to the next step.
16	**INSPECT BRAKE LIGHT (RH) FOR SHORT TO POWER SUPPLY** • Verify that BCM connector is disconnected. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off or on). • Measure the voltage at the BCM terminal 2AB (wiring harness-side). • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Replace the rear combination light (RH), then go to the next step. (See REAR COMBINATION LIGHT REMOVAL/ INSTALLATION.)
17	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the BCM using the M-MDS. (See CLEARING DTC [BCM].) • Depress the brake pedal. • Retrieve the BCM DTCs using the M-MDS. (See DTC INSPECTION [BCM].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the BCM. (See BODY CONTROL MODULE (BCM) REMOVAL/ INSTALLATION.) Go to the next step.
		No	Go to the next step.
18	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [BCM].)
		No	DTC troubleshooting completed.